CTOI 159614322.01: a third TESS transit in Sector 21 narrows a Jupiter-sized duotransit to P = 89.965 d or 179.930 d

Jon McBee (independent researcher), jon.mcbee@gmail.com · 2026-10-11 · analysis of public TESS and Gaia data

Summary

CTOI 159614322.01 was submitted in October 2021 (ExoFOP user *vogl*, "in-house single transit finder") as a single transit in Sector 41, with no period. It has no follow-up on ExoFOP. We find two more transits in archival TESS data:

• **Sector 14** (BTJD 1703.828): a full transit of the same depth and duration, in the TESS-SPOC and QLP FFI light curves. It leaves P = 719.72/n.

• **Sector 21** (BTJD 1883.774): ingress and about 9 h of the transit bottom, up to the orbit-1 downlink gap. The egress is not covered. Our automated pipeline therefore left this window unclassified, and we examined it by hand.

Periods. Flat TESS data exclude every n except 1, 2, 4, 5 and 8. The S21 event falls exactly where n = 4 and n = 8 predict a transit, and at phase 0.25 or 0.5 of the others.

If S21 is a transit, the period is **89.965 d or 179.930 d**.

The companion. Its radius is about 0.91 R_{Jup}, and the host is a Gaia RV-stable G subgiant (Teff 5885 K, R* 1.65 R_{sun}).

Next test. Both periods predict a transit on **2026-12-28 at 06:42 UTC ± 45 min**. The formal error is ± 21 min, inflated ×2.1; see the ephemeris note. The target is circumpolar from mid-northern latitudes.

Ephemeris (BJD_TDB)	P = 89.965 d (n = 8)	P = 179.930 d (n = 4)
P (d)	89.9651 ± 0.0006	179.9302 ± 0.0012
T0	2458703.8277 ± 0.0041	2458703.8277 ± 0.0041
Cycles of S14 / S21 / S41	0 / 2 / 8	0 / 1 / 4
S21 O–C (min)	+23 / +13 / +26 (TESS-SPOC SAP / QLP / TESS-SPOC PDC)	same

How the ephemeris was made. *P and T0 come from the two full transits (S14, S41) only. The partial S21 transit is used only to select n = 4 or 8, and as a consistency check. Transit times (BTJD), each fitted with one transit shape plus a local linear baseline: S14 1703.8277 ± 0.0041 (TESS-SPOC PDC); S21 1883.7739 ± 0.0069 (TESS-SPOC SAP); S41 2423.5483 ± 0.0025 (SPOC_120 PDC). Against the S14–S41 ephemeris, S21 arrives +13 to +26 min late (1.2–2.1σ), possibly because it is a partial transit. A fit through all three transits has $\chi^2 = 4.5$ for 1 dof. To allow for underestimated timing errors, predicted transit times carry a planning uncertainty of the formal error × $\sqrt{\chi^2}$ = ×2.1.*

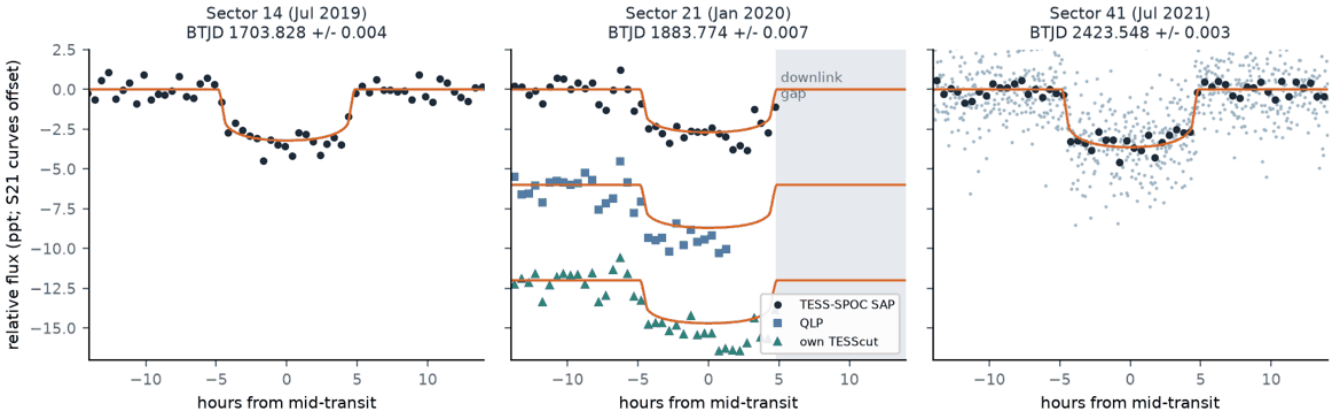


Figure 1. The three transits, with one transit shape (orange) scaled to each event's fitted depth. Sector 21 is shown in three reductions: TESS-SPOC SAP, QLP, and our own background-subtracted TESScut photometry (2-px radius). The data end at the downlink gap just before the predicted egress.

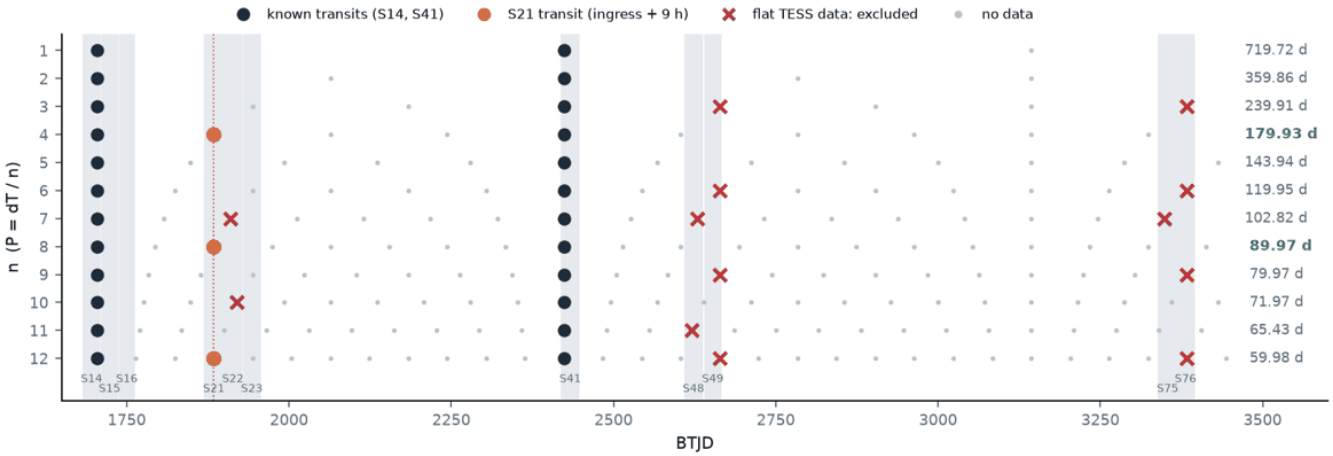


Figure 2. Every period P = 719.72/n (n = 1–12) against TESS coverage (grey: sectors observed). A red cross marks a predicted transit on flat data, which excludes that n. Only n = 1, 2, 4, 5 and 8 survive. n = 4 and 8 predict the S21 event; n = 1, 2 and 5 do not.

Is the Sector 21 event real?

Check	Result
Independent reductions	TESS-SPOC SAP: depth scale 0.76 ± 0.11. QLP: 0.95 ± 0.14. Own TESScut photometry with a local background: 4.3, 3.3, 3.5, 3.2, 2.8 and 2.7 ppt for apertures of radius 1–4 px (4–52 pixels). Each is ≥ 8σ; S14/S41 depth is about 3.3 ppt.
Timing and shape	Sharp ingress at BTJD ≈ 1883.57, which n = 4/8 predict at 1883.58. $\Delta\chi^2$ (transit vs. no transit, same baseline freedom): 60 (TESS-SPOC SAP), 52 (QLP); against a quadratic baseline 54 and 21.
Background / scattered light	The TESScut background is flat at 74.2–74.9 e/s/px from the ingress to 1883.89. It rises (scattered light) only in the last 4 cadences before the gap.
Pointing	No step at the ingress. The centroid drifts smoothly by < 0.01 px across the window.
Systematics at orbit ends	The same test at the end of all 58 orbits observed for this star (all products) finds S21 the largest by far. The largest drop at any other orbit end is 1007 ppm (S75 QLP, 4.6σ).
Source location	Transit-amplitude maps (Fig. 3) put the source on the target in S14 (2.2" ± 2.7") and S41 (1.1" ± 1.2"). In S21 the fit gives 13.5" ± 3.6" (SNR 5.5; 43 frames, no post-egress baseline). No catalogued star can host the signal there: making 3 ppt needs G < 17.9, and the nearest Gaia star that bright is 39.8" away (G = 14.5). The aperture test (depth largest in the smallest aperture) also points at the target.
Blind re-analysis	A separate agent re-analysed the star with fresh code, given only the S14 and S41 transits (independent/report.md). It found the S21 dip independently and narrowed P to the same two values. It measured Tc3 = 1883.767 ± 0.007 and depths of 3.3–3.6 ppt in SAP, QLP and its own photometry. Sliding the transit model over all 462 product-days, S21 is the only dip seen in two products. Its verdict: probably a real, on-target transit; the claim is 'narrowed to two periods', not 'resolved'.

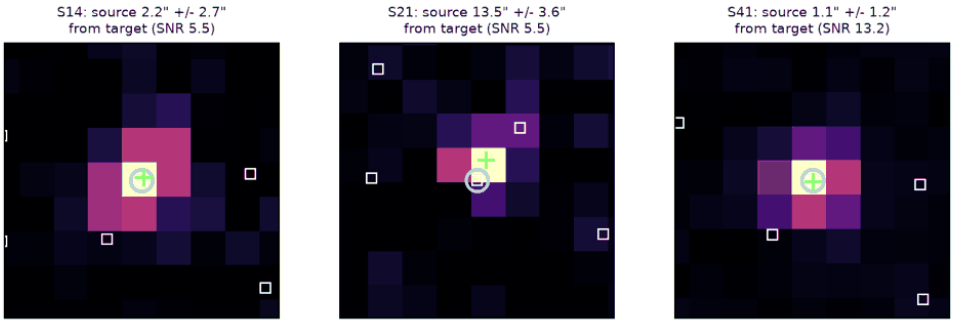


Figure 3. Per-pixel transit-amplitude maps (each pixel: quadratic baseline + transit shape). Blue circle: target; green cross: fitted source position; white squares: Gaia stars with G < 16.

Star and companion

Quantity	Value	Source
TIC / Gaia DR3	159614322 / 1669046625055549312	ExoFOP, Gaia DR3
RA, Dec (J2000)	220.00977, +65.48404	TIC
Tmag / G / BP–RP	11.18 / 11.63 / 0.87	TIC, Gaia DR3

Distance	401 pc (parallax 2.495 ± 0.019 mas)	Gaia DR3
Teff / log g / [M/H]	5885 K / 4.11 / +0.26 (GSP-Phot)	Gaia DR3
R* / M* / age	1.65 R_{sun} / 1.11 M_{sun} / 7.0 Gyr ($\rho^* \approx 0.35$ g cm^{-3})	Gaia DR3 FLAME
Binarity	RUWE 1.51 (mildly elevated); ipd_frac_multi_peak 0; non_single_star 0; RV -59.5 ± 0.4 km/s, not variable (rv_chisq_pvalue 0.88, 22 transits)	Gaia DR3
Companion	$R_p/R^* = 0.0551$, so $R_p \approx 0.91\ R_{Jup}$; T14 ≈ 9.7 h; depth ≈ 3.3 ppt	this work

Vetting of the S14 and S41 transits

Check	Result
Depth consistency	Per-epoch depth scale S14 0.99 ± 0.05, S41 1.00 ± 0.03 (χ^2 0.07/1)
Products	Depth scale: QLP SAP 1.07, TESS-SPOC PDC 1.05, TESS-SPOC SAP 0.99, SPOC_120 PDC 1.04, SPOC_120 SAP 1.03. A separate detrending code re-detects both events at SNR 19.
Centroids (SPOC)	No in-transit shift: 0.1σ and 0.2σ
Data quality	No momentum dumps or segment edges near either transit
Secondary eclipse	P = 89.965 d: -5 ± 78 ppm at phase 0.5 (S49, S75). P = 179.930 d: phase 0.5 never observed.
Transit shape vs. star	Circular orbit, b = 0: $\rho_{circ} = 0.98$ g cm^{-3} at P = 89.97 d. The FLAME density (0.35) needs b ≈ 0.7 ($\Delta\chi^2 \approx 5$) or e of at least ~0.3. At 179.93 d it needs b ≈ 0.85 or e of at least ~0.45. So the duration mildly favours the shorter period for a circular orbit; eccentricity relaxes this.
Catalogues	Not a TOI; not in the TESS EB catalogue (Prša et al. 2022); no confirmed planet; ExoFOP has no imaging, spectra or notes. A web search found no other mention of this TIC.

Predicted transits

Mid-transit (UTC)	σ plan (formal), min	Predicted by	Fraction observable* Boston / Kitt Peak / La Palma
2026-12-28 06:42	45 (21)	both periods	72% / 36% / 40%
2027-03-28 05:51	47 (22)	89.965 d only	84% / 80% / 52%
2027-06-26 05:01	49 (23)	both periods	56% / 64% / 48%
2027-09-24 04:11	50 (24)	89.965 d only	56% / 28% / 0%
2027-12-23 03:20	52 (25)	both periods	64% / 4% / 40%
2028-03-22 02:30	54 (26)	89.965 d only	72% / 48% / 84%
2028-06-20 01:40	56 (26)	both periods	48% / 32% / 68%

** Fraction of the transit ± 1 h observable with the target above 20° and the Sun below –12°. Transits on the 179.930-d ephemeris are every other one in the list. tess-point lists Sectors 120–122 and 126–128 as the next TESS visits (2027–28).*

Suggested follow-up

(1) **Photometry on 2026-12-28.** A detection confirms the S21 transit and both periods; a flat light curve would point back to the 720/360/144-d aliases. The transit is long (~9.7 h) and the window is ±45 min, so two longitudes and a ≥0.4 m telescope help. From New England the star is at only 17–20° at ingress (observing_plan_providence.md).

(2) **2027-03-28.** Only the 89.965-d period predicts a transit on this date, so it separates the two periods.

(3) **RVs.** At V ≈ 11.9, a Jupiter-mass companion at 90–180 d gives K ≈ 34–42 m/s. A stellar companion would show km/s variations.

(4) **High-resolution imaging.** Gaia RUWE is 1.51, so imaging is worth doing.

Methods: the repeat search and alias pipeline were validated blind on 26 TESS duotransit hosts with known periods (true period kept 26/26, no wrong unique answers). Code and intermediate files are available on request. Data: NASA TESS (SPOC, TESS-SPOC, QLP via MAST; TESScut) and ESA Gaia DR3. This analysis was done with substantial help from an AI system (Anthropic's Claude); the author has checked the results against the saved data and figures.